

DriveSafe-VISION-DMS-v2

Model Technical Specifications & Performance Metrics

1. MODEL ARCHITECTURE

The system utilizes a dual-stream Convolutional Neural Network (CNN) approach based on the **MobileNetV2** architecture, optimized for real-time inference on edge devices like the NVIDIA GTX 1650.

Base Model	MobileNetV2 (Pre-trained on ImageNet)
Input Size	224 × 224 × 3 (RGB)
Framework	TensorFlow 2.x / Keras
Target Hardware	NVIDIA GeForce GTX 1650 (Turing)

2. DATASET & TRAINING CONFIGURATION

Dataset Sources	MRL Eye Dataset / CEW (Closed Eyes in the Wild) / Custom Yawn Dataset
Class Labels	Binary Classification (Eye: Open/Closed Yawn: Yawn/No-Yawn)
Optimizer	Adam (Learning Rate: 1e-4)
Loss Function	Binary Cross-Entropy

Training Epochs	25-30 (with Early Stopping)
Batch Size	32

3. FINAL PERFORMANCE METRICS

The following metrics represent the validation performance achieved during the training phase (Tesla T4 Environment).

98.2%

EYE STATE ACCURACY

96.5%

YAWN DETECTION ACCURACY

~14ms

INFERENCE LATENCY (GPU)

0.97

F1-SCORE

4. DEPLOYMENT OPTIMIZATION

To improve deployment efficiency, the system was optimized using **frame sampling and skip logic**, **reduced-resolution face detection**, and **ROI-based inference** for the eye and mouth regions instead of full-frame processing.

Temporal filtering with blink handling and alarm cooldown was retained to preserve detection reliability while reducing false alerts, and **TensorFlow GPU memory control** was added for more stable runtime behavior. In addition, the application was **containerized using Docker** with a CUDA-enabled base image to support reproducible and deployment-ready execution.